The `luatests` Package in LaTeX

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1 Introduction

The `luatests` package is developed to perform various statistical tests directly within LaTeX documents. It leverages Lua's computational capabilities to calculate statistical metrics and format the results in LaTeX tables. The package provides functions for Welch's T-Test, Independent T-Test, Paired T-Test, One-Sample T-Test, and Two-Way ANOVA. These commands can be used in an environment similar to a `tabular` or `array` environment and can be formatted using optional arguments.

The package is written in Lua and requires the LuaLaTeX engine for compilation. It uses the `luacode` package to embed Lua code within LaTeX documents. The package is released under the LaTeX Project Public License v1.3c or later.

2 Installation and License

To install the `luatests` package, place the `luatests.sty` file in your LaTeX directory. Include the package in your document with:

```
\usepackage{luatests}
```

The package is released under the LaTeX Project Public License v1.3c or later. The complete license text is available at `http://www.latex-project.org/lppl.txt`.

3 Commands in the `luatests` Package

The `luatests` package provides the following commands:

- `\luaWelchsTTest`: Welch's T-Test
- `\luaTTestIndependent`: Independent T-Test (Equal Variance Assumption)
- `\luaTTestPaired`: Paired T-Test

- `\luaOneSampleTTest`: One-Sample T-Test
- `\luaOneWayAnova`: One-Way ANOVA
- `\luaTwoWayAnova`: Two-Way ANOVA

Each command has optional arguments for formatting the output table, such as `label`, `headline`, and `midline`.

3.1 Welch's T-Test

3.1.1 Description

Welch's T-Test is used to compare the means of two groups when the variances are not assumed to be equal.

3.1.2 Syntax

```
\luaWelchsTTest[options]{array1}{array2}
```

3.1.3 Options

- `label`: Custom label for the table (default: "T-Test").
- `headline`: Custom headline for the table.
- `midline`: Custom midline for the table.

3.1.4 Example

```
\begin{tabular}{c|c}
\luaWelchsTTest[label=Welch's T-Test, headline=\hline, midline=\hline]
{1, 2, 3, 4}{5, 6, 7, 8}
\end{tabular}
```

Welch's T-Test	Value
T-statistic	-4.38178
Mean of Group 1	2.5
Mean of Group 2	6.5
Variance of Group 1	1.66667
Variance of Group 2	1.66667
Standard Error	0.91287
Degrees of Freedom	6.0

The output will be a LaTeX table with the T-statistic, means, variances, standard error, and degrees of freedom.

3.2 Independent T-Test (Equal Variance Assumption)

3.2.1 Description

This test compares the means of two independent groups assuming equal variances.

3.2.2 Syntax

```
\luaTTestIndependent[options]{array1}{array2}
```

3.2.3 Options

- `label`: Custom label for the table (default: "T-Test").
- `headline`: Custom headline for the table.
- `midline`: Custom midline for the table.

3.2.4 Example

```
\luaTTestIndependent[label=Independent T-Test, headline=\hline, midline=\hline]
{1, 2, 3, 4}{5, 6, 7, 8}
```

Independent T-Test	Value
T-statistic	-4.38178
Mean of Group 1	2.5
Mean of Group 2	6.5
Variance of Group 1	1.66667
Variance of Group 2	1.66667
Standard Error	0.91287
Degrees of Freedom	6.0

The output will be a LaTeX table with the T-statistic, means, variances, standard error, and degrees of freedom.

3.3 Paired T-Test

3.3.1 Description

This test compares the means of two related groups (e.g., before and after measurements).

3.3.2 Syntax

```
\luaTTestPaired[options]{array1}{array2}
```

3.3.3 Options

- `label`: Custom label for the table (default: "T-Test").
- `headline`: Custom headline for the table.
- `midline`: Custom midline for the table.

3.3.4 Example

```
\begin{tabular}{c|c}
    \luaTTestPaired[label=Paired T-Test, headline=\hline, midline=\hline]
    {1, 2, 3, 4}{5, 6, 7, 8}
\end{tabular}
```

Paired T-Test	Value
T-statistic	$-inf$
Mean Diff	-4.0
Variance Diff	0.0
Standard Error	0.0
Degrees of Freedom	3.0

The output will be a LaTeX table with the T-statistic, mean difference, variance difference, standard error, and degrees of freedom.

3.4 One-Sample T-Test

3.4.1 Description

This test compares the mean of a single group to a known population mean.

3.4.2 Syntax

```
\luaOneSampleTTest[options]{array}{population_mean}
```

3.4.3 Options

- `label`: Custom label for the table (default: "T-Test").
- `headline`: Custom headline for the table.
- `midline`: Custom midline for the table.

3.4.4 Example

```
\begin{tabular}{c|c}
 \luaOneSampleTTest[label=One-Sample T-Test, headline=\hline, midline=\hline]
 {1, 2, 3, 4}{2.5}
\end{tabular}
```

One-Sample T-Test	Value
T-statistic	0.0
Mean	2.5
Variance	1.66667
Standard Error	0.6455
Degree of Freedom	3

The output will be a LaTeX table with the T-statistic, mean, variance, standard error, and degrees of freedom.

3.5 One-Way ANOVA

3.5.1 Description

This test analyzes the effect of two independent categorical variables on a continuous outcome.

3.5.2 Syntax

```
\luaOneWayAnova[options]{data}
```

3.5.3 Options

- `headline`: Custom headline for the table.
- `midline`: Custom midline for the table.

3.5.4 Example

```
\begin{tabular}{c|c|c|c|c}
\luaOneWayAnova[headline=\hline, midline=\hline]{{1, 2, 3}, {4, 5, 6}, {7, 8, 9}}
\end{tabular}
```

Source of Variation	**Sum of Squares**	**df**	**Mean Square**	F_{value}
Between Groups	91.16667	2	45.58333	42.07692
Within Groups	9.75	9	1.08333	
Total	100.91667	11		

The output will be a LaTeX table with the sum of squares, degrees of freedom, mean squares, and F-values for both factors and the error term.

3.6 Two-Way ANOVA

3.6.1 Description

This test analyzes the effect of two independent categorical variables on a continuous outcome.

3.6.2 Syntax

```
\luaTwoWayAnova[options]{data}
```

3.6.3 Options

- `headline`: Custom headline for the table.
- `midline`: Custom midline for the table.

3.6.4 Example

```
\begin{tabular}{c|c|c|c|c}
\luaTwoWayAnova[headline=\hline, midline=\hline]{{1, 2, 3}, {4, 5, 6}, {7, 8, 9}}
\end{tabular}
```

Source of Variation	**Sum of Squares**	**df**	**Mean Square**	F_{value}
Factor A	91.16667	2	45.58333	78.14286
Factor B	6.25	3	2.08333	3.57143
Error	3.5	6	0.58333	
Total	100.91667	11		

The output will be a LaTeX table with the sum of squares, degrees of freedom, mean squares, and F-values for both factors and the error term.

Conclusion

The `luatests` package provides a powerful way to perform statistical tests directly within LaTeX documents. By leveraging Lua's computational capabilities, it allows for dynamic calculations and formatted output, making it ideal for academic and scientific documents.

More Examples

Example 1: Welch's T-Test with Headline

T-Test	Value
T-statistic	-2.4057
Mean of Group 1	5.0
Mean of Group 2	8.5
Variance of Group 1	9.2
Variance of Group 2	3.5
Standard Error	1.45488
Degrees of Freedom	8.32336

Example 2: Welch's T-Test with Row Color

T-Test	Value
T-statistic	-2.4057
Mean of Group 1	5.0
Mean of Group 2	8.5
Variance of Group 1	9.2
Variance of Group 2	3.5
Standard Error	1.45488
Degrees of Freedom	8.32336

Example 3: Independent T-Test with Headline

T-Test	Value
T-statistic	-2.4057
Mean of Group 1	5.0
Mean of Group 2	8.5
Variance of Group 1	9.2
Variance of Group 2	3.5
Standard Error	1.45488
Degrees of Freedom	10.0

Example 4: Independent T-Test with Midline

T-Test	Value
T-statistic	-2.4057
Mean of Group 1	5.0
Mean of Group 2	8.5
Variance of Group 1	9.2
Variance of Group 2	3.5
Standard Error	1.45488
Degrees of Freedom	10.0

Example 5: Paired T-Test with Both Headline and Midline

T-Test	Value
T-statistic	-7.0
Mean Diff	-3.5
Variance Diff	1.5
Standard Error	0.5
Degrees of Freedom	5.0

Example 6: One-Sample T-Test with Dataset

T-Test	Value
T-statistic	3.06683
Mean	21.4
Variance	6.46008
Standard Error	0.4565
Degree of Freedom	30

Example 7: Two-Way ANOVA with Row Color

Source of Variation	**Sum of Squares**	**df**	**Mean Square**	F_{value}
Factor A	6.16667	2	3.08333	1.21978
Factor B	3.33333	3	1.11111	0.43956
Error	15.16667	6	2.52778	
Total	24.66667	11		

Example 8: One-Way ANOVA with Headline and Midline

Source of Variation	**Sum of Squares**	**df**	**Mean Square**	F_{value}
Between Groups	6.16667	2	3.08333	1.5
Within Groups	18.5	9	2.05556	
Total	24.66667	11		